The W-algebra of a polynomial functor is homotopy-initial

The Univalent Foundations Program

hott-book:induction.tex:local-snapshot-20261012; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's labelled theorem thm:w-hinit at source lines 524-526; its proof environment starts on line 528, the line immediately after the blank line that follows the statement environment, so the pairing is by adjacency and the retained proof range 528-562 is the whole environment.

Editorial note. The two context spans retain the definition of the type of W-homomorphisms (source lines 506-510) and the definition of homotopy-initiality (source lines 512-520), because the counted statement is stated entirely in those terms. Both contain only unnumbered displays, so neither consumes a counter value.

Editorial note. The numbered display inside the retained proof is the author's own labelled equation of the proof; because the book lets the equation counter share the theorem counter, the retained counters make the counted theorem print as Theorem 5.4.7 and that display print as (5.4.8), exactly as in the book.

Editorial note. The retained proof is the author's sketch, kept as such; nothing is added to complete it and the two pasting diagrams are reproduced from the same pinned source.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-source copy at sources/hott-induction/retained-preamble.tex, referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the W-algebra of a polynomial functor is homotopy-initial (author label thm:w-hinit, original Theorem 5.4.7).

Allowed standard prerequisites (editorial).

- W-types. For A a type and B a family over A, the W-type of B has a constructor taking a : A and t : B(a) to W and satisfies an elimination rule with a computation rule.. The introduction rule gives the structure map at source line 532 and the elimination and computation rules give the homomorphism at source lines 536-537.

- Polynomial functor and algebra. The polynomial functor of A and B sends X to the type of pairs (a,t) with a : A and t : B(a) to X, and an algebra is a type with a map from its functor image.. It is the functor P and the algebras (W,s_W) and (C,s_C) of source lines 530-535.
- Algebra homomorphism and 2-cell. A homomorphism is a map together with a homotopy witnessing commutation with the structure maps, and a 2-cell between homomorphisms is a pair of a homotopy between the maps and a higher witness between the two pasting diagrams.. It is the reduction of an identity of homomorphisms performed at source lines 543-558.
- Function extensionality. A pointwise homotopy between two functions gives an equality of functions.. It is named at source line 559 as the tool that produces the path between the two homomorphisms.

Source reference. Original section 5.3: sources/hott-induction/book-labelnumbers-first-edition.tex, source lines 1724–1724. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

We have the type of W-homomorphisms:

$$\mathsf{WHom}_{A,B}((C,s_C),(D,s_D)) :\equiv \sum_{(f:C\to D)} \prod_{(a:A)} \prod_{(h:B(a)\to C)} f(s_C(a,h)) = s_D(a, f\circ h)$$

Finally, a P-algebra (C,s_C) is said to be **homotopy-initial** if for every P-algebra (D,s_D), the type of all algebra homomorphisms $(C,s_C)\to(D,s_D)$ is contractible. That is,

$$\mathsf{isHinit}_{\mathsf{W}}(A,B,I) :\equiv \prod_{C:\mathsf{WAlg}(A,B)} \mathsf{isContr}(\mathsf{WHom}_{A,B}(I,C)).$$

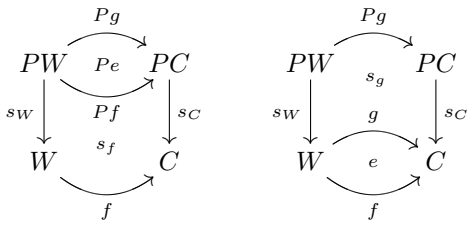
Theorem 5.4.7. *For any type $A:\mathcal{U}$ and type family $B:A\to\mathcal{U}$, the W-algebra $(\mathsf{W}_{(x:A)}B(x),\mathsf{sup})$ is h-initial.*

Sketch of proof. Suppose we have $A:\mathcal{U}$ and $B:A\to\mathcal{U}$, and consider the associated polynomial functor $P(X) :\equiv \sum_{(x:A)}(B(x)\to X)$. Let $W :\equiv \mathsf{W}_{(x:A)}B(x)$. Then using the W-introduction rule from original section 5.3, we have a structure map $s_W :\equiv \mathsf{sup} : PW\to W$. We want to show that the algebra (W,s_W) is h-initial. So, let us consider another algebra (C,s_C) and show that the type $T :\equiv \mathsf{WHom}_{A,B}((W,s_W),(C,s_C))$ of W-homomorphisms from (W,s_W) to (C,s_C) is contractible. To do so, observe that the W-elimination rule and the W-computation rule allow us to define a W-homomorphism $(f,s_f):(W,s_W)\to(C,s_C)$, thus showing that T is inhabited. It is furthermore necessary to show that for every W-homomorphism $(g,s_g):(W,s_W)\to(C,s_C)$, there is an identity proof

$$p : (f,s_f) = (g,s_g). \tag{5.4.8}$$

This uses the fact that, in general, a type of the form $(f,s_f)=(g,s_g)$ is equivalent to the type of what we call **algebra 2-cells** from f to g, whose canonical

elements are pairs of the form (e, s_e), where $e : f = g$ and s_e is a higher identity proof between the identity proofs represented by the following pasting diagrams:



In light of this fact, to prove that there exists an element as in (5.4.8), it is sufficient to show that there is an algebra 2-cell (e, s_e) from f to g. The identity proof $e : f = g$ is now constructed by function extensionality and W-elimination so as to guarantee the existence of the required identity proof s_e. □